

# From Slab to Beam

Tributary Areas, Line Loads and a Two-Dimensional Subframe

| Course | Computer (& AI) Methods for Structural Engineering (DISC 5203) |
|---|---|
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| Assigned plan | Floor Plan 1 (final digit 1) |
| Review date | 6 October 2026 |

**24 beam segments | net slab 180.128 m2 | four OpenSees models | exact area and reaction equilibrium**

## Scope

This report extends the existing HW1 Git working tree. It reuses the HW1 slab and vertical-element results, transfers the same gravity load to beams with 45-degree tributary boundaries, models the two marked beams as standalone continuous beams and one-floor subframes, and cross-checks subframe column forces against HW1 axial estimates.

# 1. Reused plan and assumptions

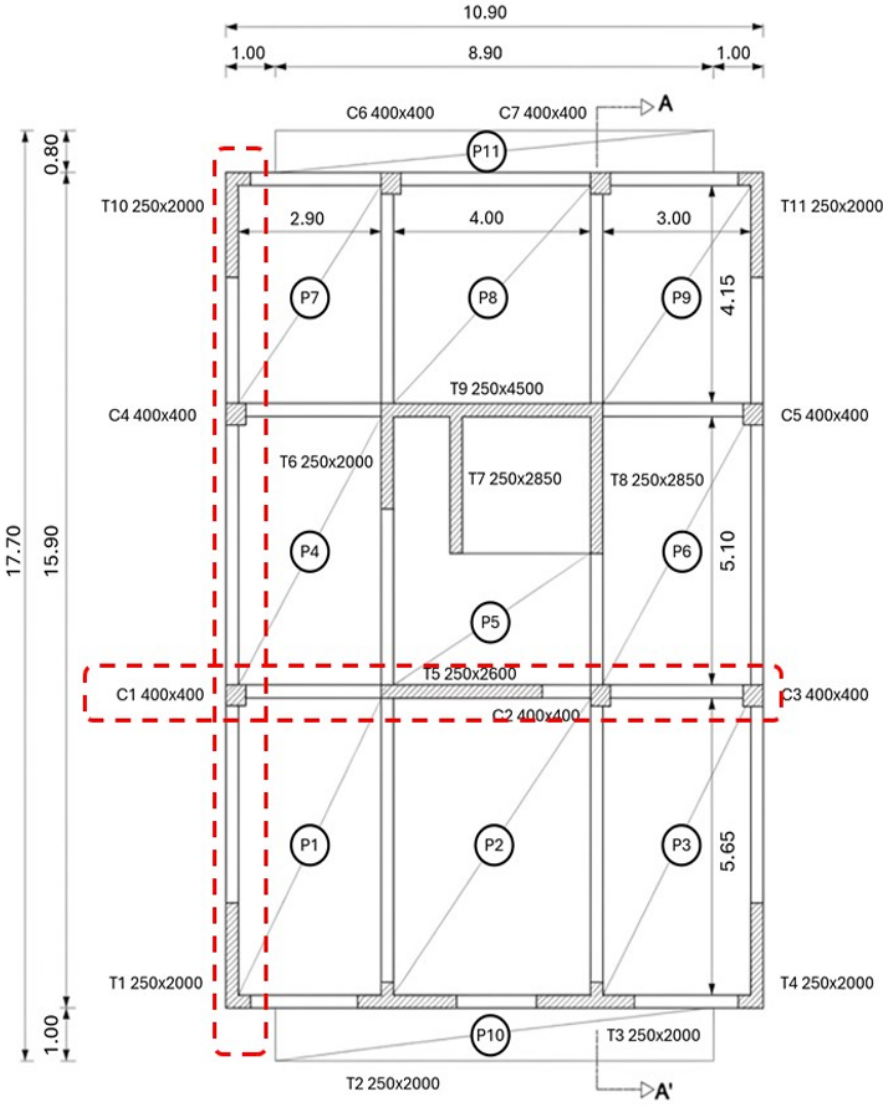



Figure 1. Assigned Floor Plan 1; red boxes mark Beam A and Beam B.

Dead slab load = 25 x 0.150 = 3.75 kN/m2; live load = 5.00 kN/m2. No finish load is printed. Beam self-weight is excluded from the requested slab-to-beam transfer. Unprinted beam size is explicitly assumed as 300 x 600 mm; columns are 400 x 400 mm, storey height 3.0 m and E = 30.0e6 kN/m2.

## 2. 45-degree beam tributary construction

Each cell of the unchanged HW1 net slab is assigned to its nearest finite beam edge. Parallel beams divide at mid-lines; equal distance to perpendicular edges gives |Delta x| = |Delta y| and a 45-degree corner line. Openings are removed before assignment. P10 and P11 are cantilevers and add a uniform component to their single support line.

For a two-way rectangle with short side a and long side b: A_short = a2/4 and A_long = a(2b-a)/4. The stored non-uniform profile, not the tabulated UDL, is used for local response.

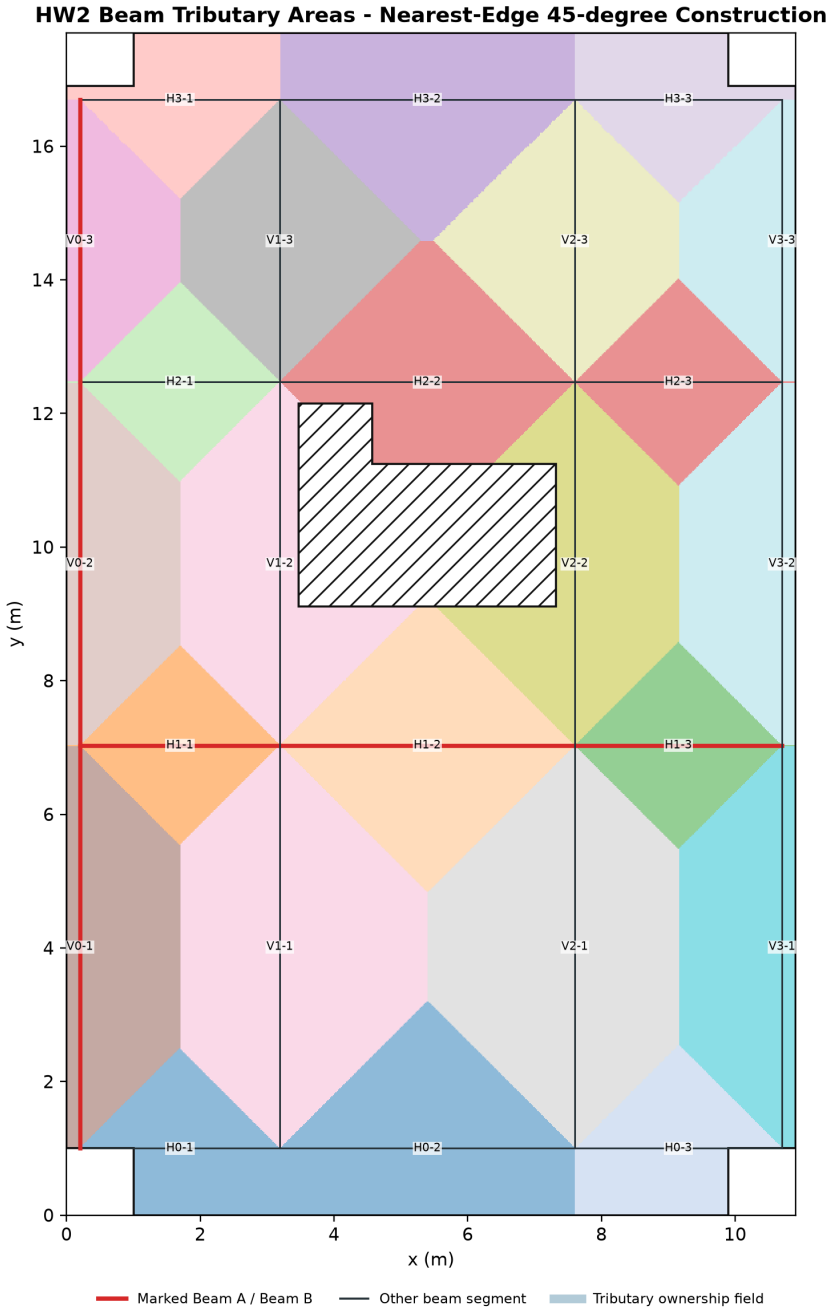



Figure 2. Beam ownership field and 45-degree divisions.

# 3. Beam census and equivalent line-load table

Equivalent UDL means equal total load only. It does not preserve local shear or the exact moment-diagram shape.

| ID | Span | Panels | Class | Area | Dead | Live | Total |
|---|---|---|---|---|---|---|---|
| H0-1 | 2.990 | P1; P10 (cantilever balcony where overlapping) | two loaded sides locally (main slab + balcony) | 4.449 | 5.580 | 7.439 | 13.019 |
| H0-2 | 4.410 | P2; P10 (cantilever balcony where overlapping) | two loaded sides locally (main slab + balcony) | 9.281 | 7.892 | 10.523 | 18.416 |
| H0-3 | 3.100 | P3; P10 (cantilever balcony where overlapping) | two loaded sides locally (main slab + balcony) | 4.706 | 5.693 | 7.591 | 13.284 |
| H1-1 | 2.990 | P1; P4 | interior, slab on both sides | 4.473 | 5.610 | 7.480 | 13.089 |
| H1-2 | 4.410 | P2; P5 | interior, slab on both sides | 9.707 | 8.254 | 11.005 | 19.259 |
| H1-3 | 3.100 | P3; P6 | interior, slab on both sides | 4.808 | 5.816 | 7.755 | 13.571 |
| H2-1 | 2.990 | P4; P7 | interior, slab on both sides | 4.473 | 5.610 | 7.480 | 13.089 |
| H2-2 | 4.410 | P5; P8 | interior, slab on both sides | 8.210 | 6.981 | 9.308 | 16.289 |
| H2-3 | 3.100 | P6; P9 | interior, slab on both sides | 4.808 | 5.816 | 7.755 | 13.571 |
| H3-1 | 2.990 | P7; P11 (cantilever balcony where overlapping) | two loaded sides locally (main slab + balcony) | 4.625 | 5.800 | 7.734 | 13.534 |
| H3-2 | 4.410 | P8; P11 (cantilever balcony where overlapping) | two loaded sides locally (main slab + balcony) | 9.247 | 7.864 | 10.485 | 18.348 |
| H3-3 | 3.100 | P9; P11 (cantilever balcony where overlapping) | two loaded sides locally (main slab + balcony) | 4.897 | 5.924 | 7.899 | 13.822 |
| V0-1 | 6.030 | P1 | perimeter, slab on one side | 7.982 | 4.964 | 6.618 | 11.582 |
| V0-2 | 5.440 | P4 | perimeter, slab on one side | 6.992 | 4.820 | 6.427 | 11.246 |
| V0-3 | 4.230 | P7 | perimeter, slab on one side | 4.947 | 4.385 | 5.847 | 10.232 |
| V1-1 | 6.030 | P1; P2 | interior, slab on both sides | 15.240 | 9.478 | 12.637 | 22.114 |
| V1-2 | 5.440 | P4; P5 | interior, slab on both sides | 8.980 | 6.190 | 8.253 | 14.443 |
| V1-3 | 4.230 | P7; P8 | interior, slab on both sides | 8.518 | 7.551 | 10.068 | 17.619 |
| V2-1 | 6.030 | P2; P3 | interior, slab on both sides | 15.383 | 9.567 | 12.756 | 22.323 |
| V2-2 | 5.440 | P5; P6 | interior, slab on both sides | 9.584 | 6.607 | 8.809 | 15.416 |
| V2-3 | 4.230 | P8; P9 | interior, slab on both sides | 8.652 | 7.670 | 10.227 | 17.896 |
| V3-1 | 6.030 | P3 | perimeter, slab on one side | 8.101 | 5.038 | 6.717 | 11.755 |
| V3-2 | 5.440 | P6 | perimeter, slab on one side | 7.088 | 4.886 | 6.515 | 11.401 |
| V3-3 | 4.230 | P9 | perimeter, slab on one side | 4.979 | 4.414 | 5.885 | 10.299 |

Units: span in m; area in m2; line loads in kN/m.

## 4. Load profiles and worked example

For H1-2, L = 4.410 m and the actual two-sided tributary area is read from P2 and P5 after the core opening is removed. D = 3.75 A, L_live = 5.00 A, and each equivalent line load is the corresponding total divided by 4.410 m. The figure retains the triangular/trapezoidal distribution used in OpenSees.

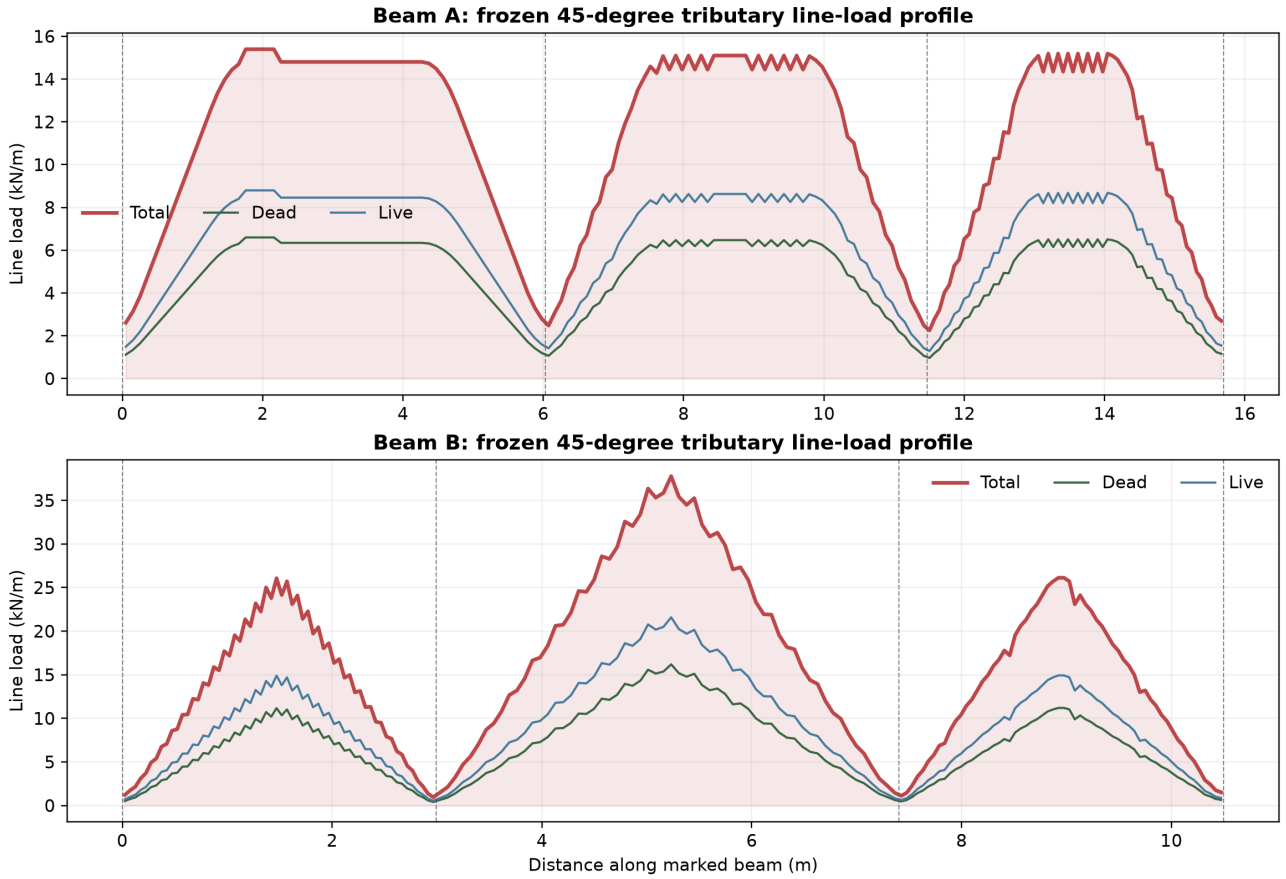



Figure 3. Frozen dead, live and total line-load profiles.

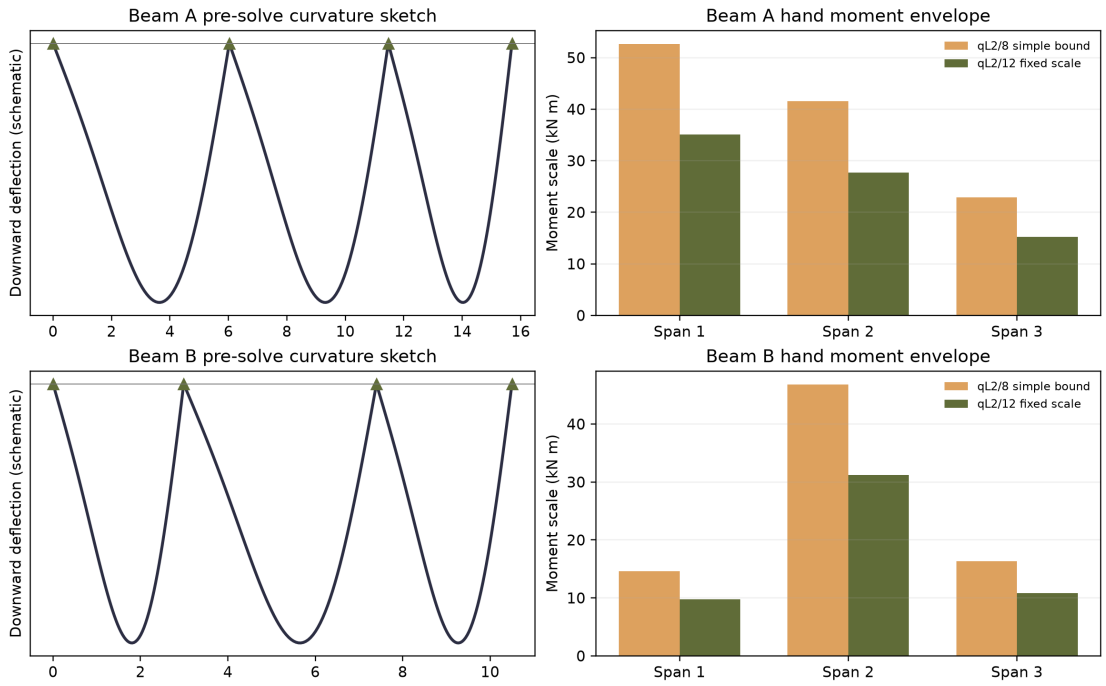



Figure 4. Pre-solve sketches and qL2 bounds.

## 5. OpenSees response diagrams

Standalone models retain vertical supports with free rotations. Subframes replace them with fixed-base 3 m columns sharing the beam nodes. The same frozen profile is applied in every comparison.

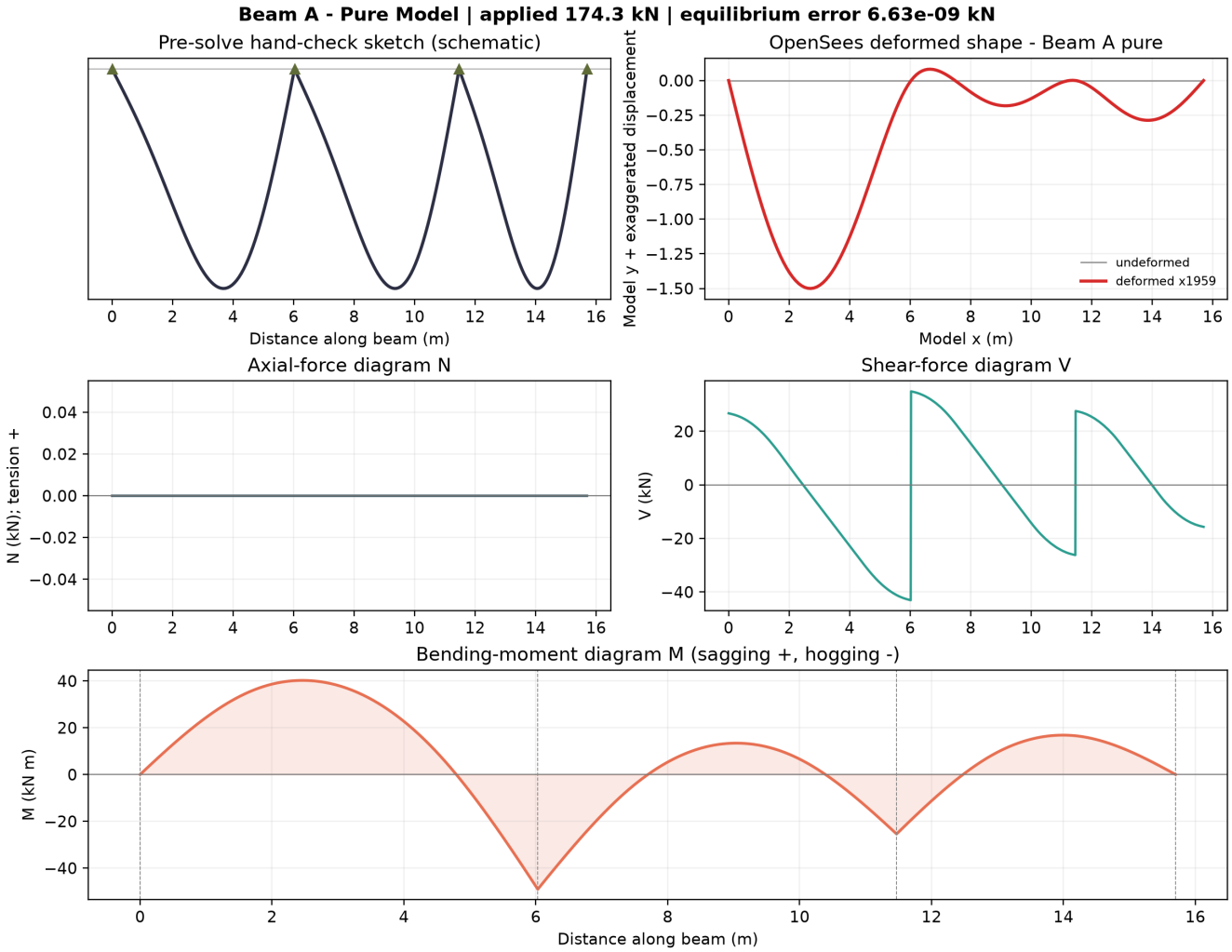



Figure. Beam A, pure model: pre-solve sketch, deformed shape and N/V/M diagrams.

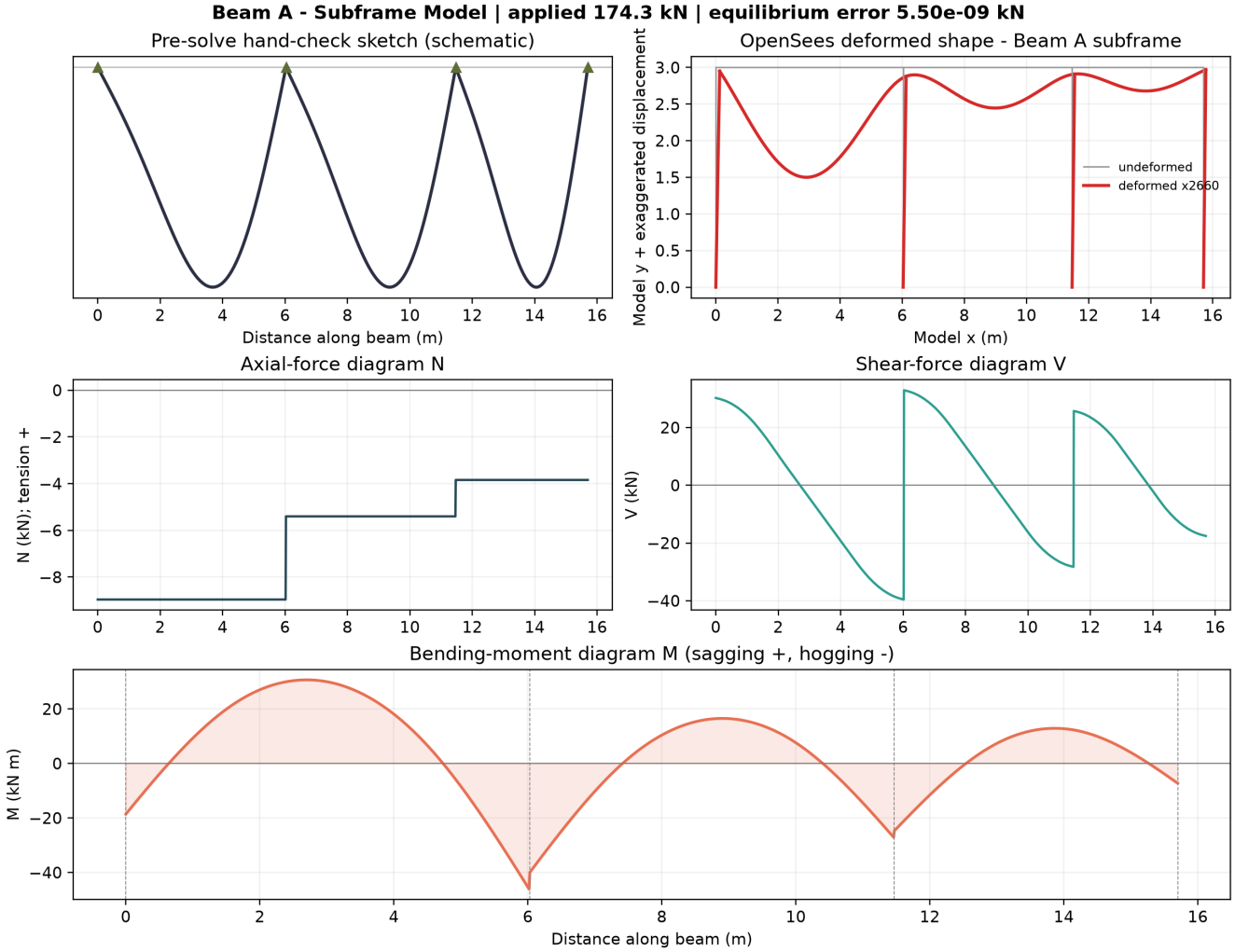



Figure. Beam A, subframe model: pre-solve sketch, deformed shape and N/V/M diagrams.

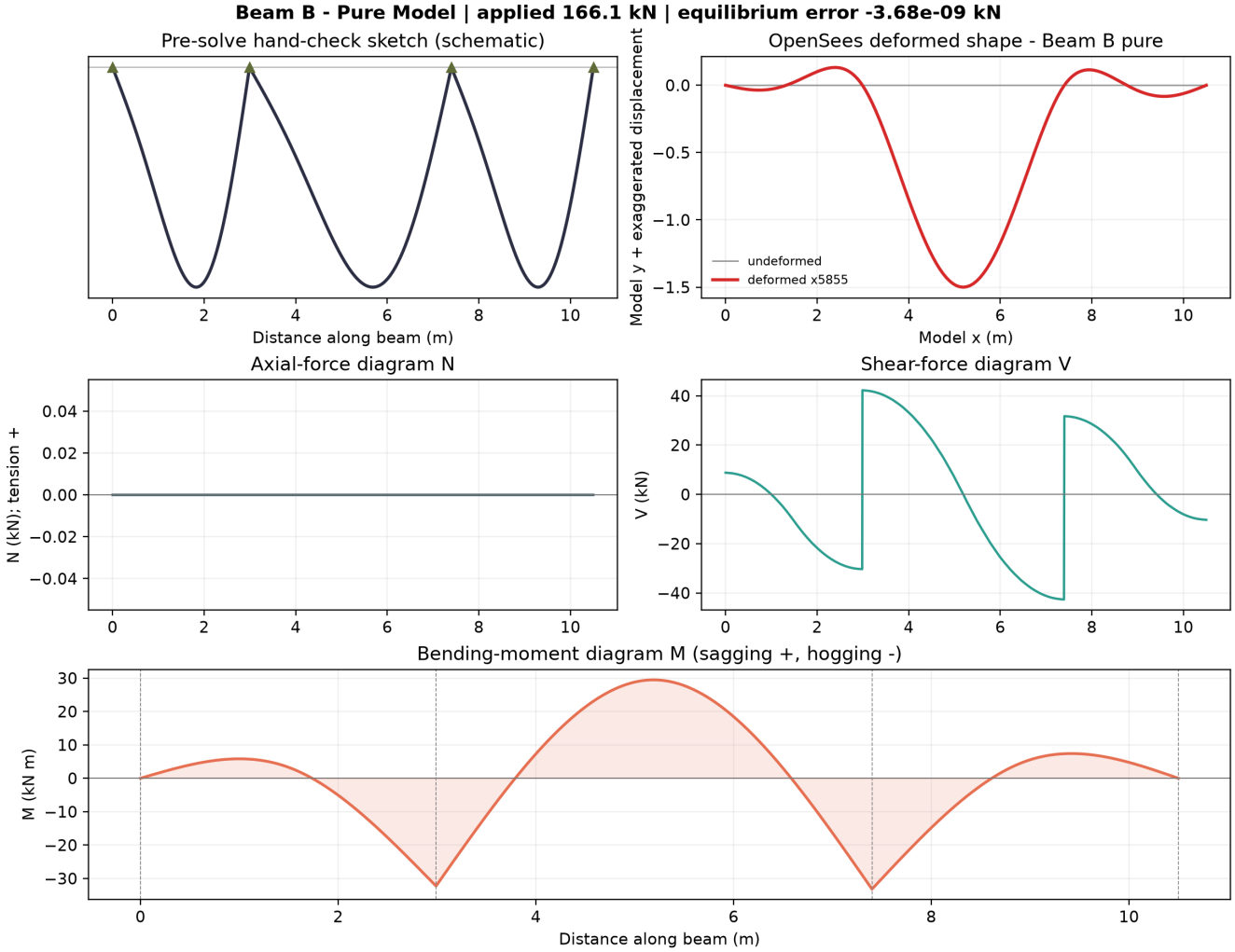



Figure. Beam B, pure model: pre-solve sketch, deformed shape and N/V/M diagrams.

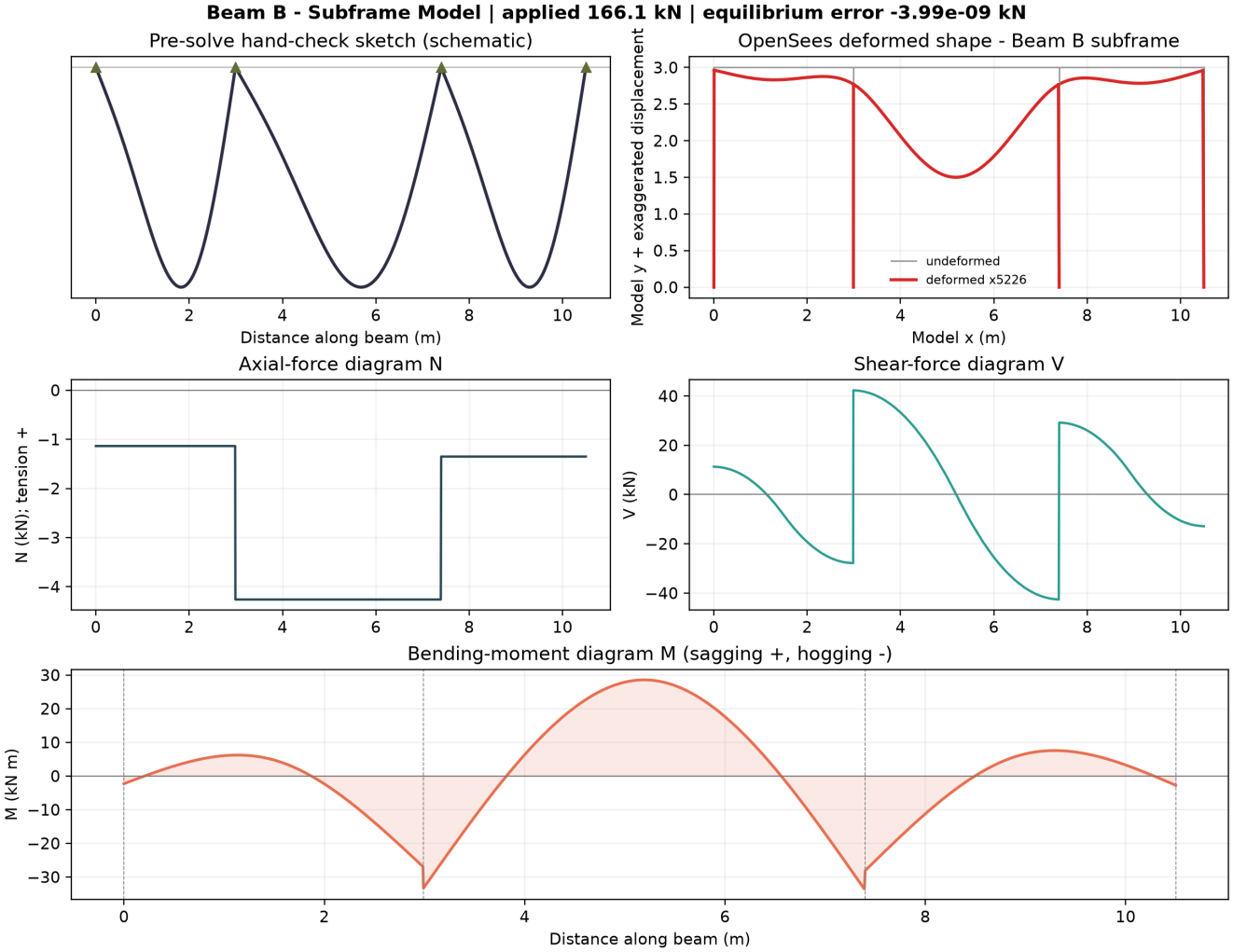



Figure. Beam B, subframe model: pre-solve sketch, deformed shape and N/V/M diagrams.

## 6. Numerical comparison

| Beam | Model | Load | Sag + | Hog - | Max \|M\| | Down mm | qL2/8 | Check |
|---|---|---|---|---|---|---|---|---|
| A | pure | 174.305 | 40.243 | -49.039 | 49.039 | 0.7656 | 52.643 | PASS |
| A | subframe | 174.305 | 30.678 | -46.096 | 46.096 | 0.5640 | 52.643 | PASS |
| B | pure | 166.139 | 29.470 | -33.191 | 33.191 | 0.2562 | 46.819 | PASS |
| B | subframe | 166.139 | 28.566 | -33.604 | 33.604 | 0.2870 | 46.819 | PASS |

Loads are kN; moments are kN m. Every model closes equilibrium below 1e-8 kN and has sagging-positive and hogging-negative regions. The subframe is the defensible beam-design model because it retains finite column stiffness; the standalone beam remains an independent bound.

## 7. HW1 axial cross-check

| Strip | Grid | HW1 member | HW1 N | HW2 N | Diff % | Sign |
|---|---|---|---|---|---|---|
| A | G00 | T1 | -111.266 | -30.239 | -72.8 | PASS |
| A | G01 | C1 | -32.699 | -72.509 | 121.7 | PASS |
| A | G02 | C4 | -42.360 | -53.999 | 27.5 | PASS |
| A | G03 | T10 | -66.258 | -17.559 | -73.5 | PASS |
| B | G01 | C1 | -32.699 | -11.302 | -65.4 | PASS |
| B | G11 | T5 | -233.992 | -70.151 | -70.0 | PASS |
| B | G21 | C2 | -36.226 | -71.814 | 98.2 | PASS |
| B | G31 | C3 | -27.321 | -12.872 | -52.9 | PASS |

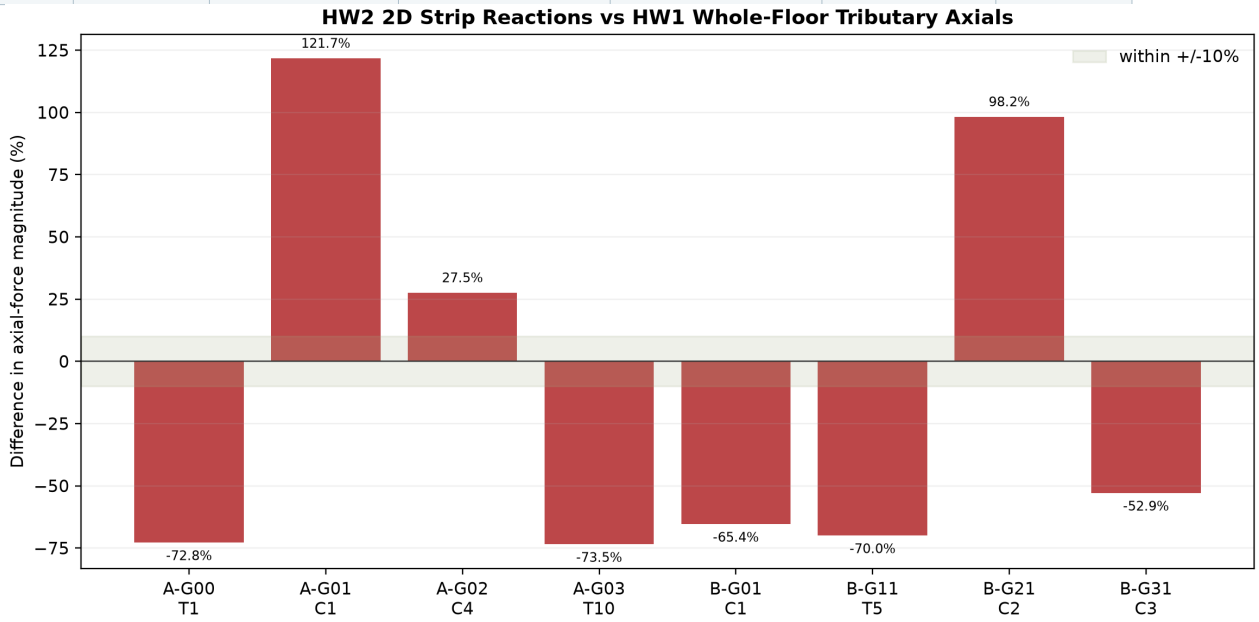



Figure. Magnitude difference between the 2D marked strips and HW1 whole-floor estimates.

Compression sign agrees at every mapped support, but the magnitude is not within a few percent. HW1 includes the whole-floor nearest-support load and aggregated walls; each HW2 result contains only one marked strip and omits perpendicular beam paths. No load was tuned. The subframe is used for the beam design question; exact cross-model reconciliation requires a 3D grid or common nodal aggregation.

## 8. Agent workflow and reproducibility

The word briefings were frozen before code generation. The agent transcribed them into four OpenSees entry files plus an audited shared implementation. Gates were: identify the deck, freeze loads, check units and signs, verify qL2 bounds, replace supports by columns, close equilibrium, compare HW1 axials, then verify the independent HW2 repository before upload.

Run: analyze_hw2_beam_loads.py; run_hw2_models.py; summarize_hw2_models.py; build_hw2_report.py; render_hw2_report_pdf.py; verify_hw2_outputs.py.

### Limitations

The plan is raster; beam size and uncracked stiffness are assumptions; the subframes are 2D strips; and the student must add a literal handwritten/signature sheet if required. These limits are preserved rather than concealed.

## Conclusion

All required beam census, tributary areas, line loads, four OpenSees cases, N/V/M and deformation diagrams, reactions, hand bounds, prompt record, HW1 cross-check and reproducibility files are present in the independent HW2 folder; the accepted HW1 snapshot remains separate and unchanged.